

1 Radial Schrödinger

We first verify the accuracy of the spectral finite-element framework for the all-electron radial Schrödinger equation, which is obtained by setting $\hat{V}_{xc} = \tilde{V}_H = 0$ in the radial Kohn–Sham equation and admits analytical solutions. With a pure Coulomb nucleus $V_{\text{nuc}}(r) = -Z/r$ and Hartree and exchange–correlation terms omitted, the radial eigenvalue problem is

$$\left(-\frac{1}{2}\frac{d^2}{dr^2} - \frac{Z}{r} + \frac{l(l+1)}{2r^2}\right)\tilde{R}_{nl} = \lambda_{nl}\tilde{R}_{nl}, \quad (1)$$

with hydrogenic levels $\varepsilon_n = -Z^2/(2n^2)$ Ha. We consider $Z = 92$ and compute all occupied eigenvalues using a domain size of $R_{\text{max}} = 40$ Bohr, an exponential mesh with shift $c = 0$ and concentration parameter $a = 100$, $N_{fe} = 12$ finite elements, polynomial degree $p = 20$, and quadrature order $q = 60$. Figure 1 shows the eigenvalue errors relative to the analytical values, demonstrating accuracies better than 10^{-10} Ha and confirming the accuracy of the spectral finite-element framework. The finite-element eigenvalues λ_{nl} and their comparison to ε_n are listed in Table 1.

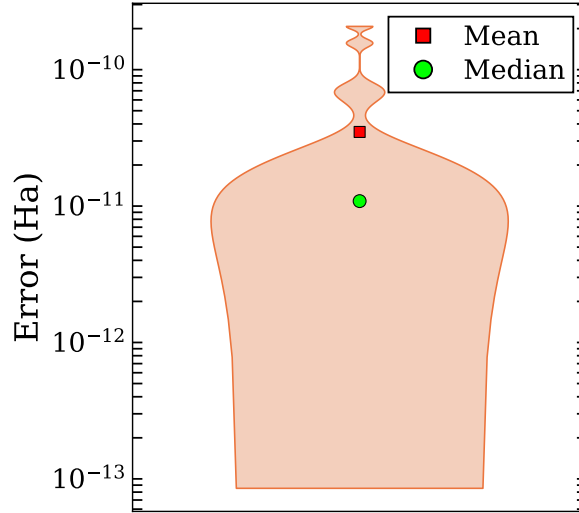


Figure 1: Distribution of eigenvalue errors for the radial Schrödinger equation corresponding to $Z = 92$.

Table 1: Occupied subshell eigenvalues for $Z = 92$ (Ha); analytic $\varepsilon_n = -Z^2/(2n^2)$.

n	l	g_{nl}	numeric (Ha)	analytic (Ha)	Error (Ha)
1	0	2	-4231.9999999998436	-4232.0000000000000	1.56×10^{-10}
2	0	2	-1057.9999999997924	-1058.0000000000000	2.08×10^{-10}
2	1	6	-1058.0000000000209	-1058.0000000000000	2.09×10^{-11}
3	0	2	-470.2222222222873	-470.2222222222222	6.50×10^{-11}
3	1	6	-470.2222222221507	-470.2222222222222	7.15×10^{-11}
3	2	10	-470.222222222160	-470.2222222222222	6.20×10^{-12}
4	0	2	-264.5000000000234	-264.5000000000000	2.34×10^{-11}
4	1	6	-264.5000000000005	-264.5000000000000	5.12×10^{-13}
4	2	10	-264.5000000000092	-264.5000000000000	9.15×10^{-12}
4	3	14	-264.4999999999808	-264.5000000000000	1.92×10^{-11}
5	0	2	-169.2800000000126	-169.2800000000000	1.26×10^{-11}
5	1	6	-169.2799999999855	-169.2800000000000	1.45×10^{-11}
5	2	10	-169.2800000000001	-169.2800000000000	8.53×10^{-14}
5	3	14	-169.2799999999980	-169.2800000000000	2.05×10^{-12}
6	0	2	-117.5555555555597	-117.5555555555556	4.11×10^{-12}
6	1	6	-117.5555555555492	-117.5555555555556	6.35×10^{-12}
6	2	10	-117.5555555555598	-117.5555555555556	4.21×10^{-12}
7	0	2	-86.3673469387809	-86.3673469387755	5.37×10^{-12}

2 All-electron Kohn–Sham discretization convergence

We now verify the accuracy and performance of SPARC-atomSFE for all-electron Kohn–Sham DFT calculations. Unless otherwise specified, we use an exponential mesh with shift $c = 0$, concentration parameter $a = 100$, domain size $R_{max} = 40$ Bohr, $N_{fe} = 12$ finite elements, polynomial degree $p = 20$, and quadrature order $q = 60$.

We begin by studying convergence with respect to the domain size R_{max} and the number of finite elements N_{fe} . In particular, Figure 2 shows the convergence of the total energy and occupied eigenvalues for the PBE functional across atoms with atomic numbers $Z = 1$ –92, with reference results computed using $R_{max} = 40$ Bohr and $N_{fe} = 16$. We observe that both quantities converge exponentially with increasing R_{max} , with errors below 10^{-6} Ha achieved at $R_{max} \sim 27$ Bohr. We also observe that there is rapid convergence with increasing N_{fe} , with accuracies better than 10^{-6} Ha attained at $N_{fe} \sim 8$. These results are consistent with previous LDA results obtained using FeAtom; however, GGA requires a somewhat larger number of degrees of freedom, with LDA achieving approximately an order of magnitude higher accuracy for comparable discretization parameters. We also find similar convergence behavior for all other exchange-correlation functionals considered here, except RPA-OEP, which exhibits slower convergence with respect to the discretization parameters.

As supplementary material to the convergence discussion presented in the main paper, the convergence sweeps are constructed as *nested* subsets of a common reference exponential mesh, rather than by regenerating independent meshes for each choice of R_{max} or N_{fe} . This construction isolates the effect of radial resolution while preserving the underlying exponential mesh distribution across discretization levels, thereby avoiding simultaneous changes in both node count and global mesh distribution during the sweep. Figure 3 illustrates this subset construction for the all-electron case.

We next compare LDA-SVWN sweep outputs for neutral uranium ($Z = 92$) to independent FeAtom occupied eigenvalues at $R_{max} = 40$ Bohr and $N_{fe} = 16$. Figure 4 shows the per-state absolute eigenvalue error as a function of R_{max} and of N_{fe} .

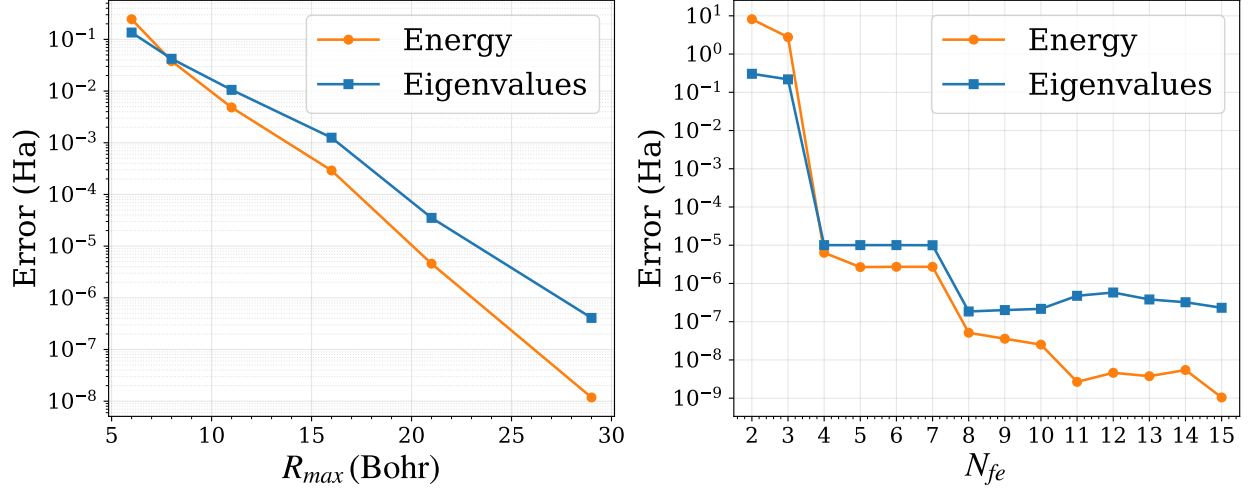


Figure 2: Convergence of the total energy and occupied eigenvalues with respect to domain size R_{max} (left) and number of finite elements N_{fe} (right) in all-electron Kohn–Sham DFT calculations, using the PBE exchange–correlation functional, for atomic numbers $Z = 1$ –92. The energy error is defined as the maximum absolute total-energy difference over all atoms, while the eigenvalue error is defined as the maximum, over all atoms, of the per-atom mean absolute difference in the occupied eigenvalues. Reference values are computed using domain size $R_{max} = 40$ Bohr and $N_{fe} = 16$ finite elements.

Finally, Figures 5, 6, and 7 give LDA–SVWN, rSCAN, and Hartree–Fock counterparts to Figure 2: the same two-panel layout, error definitions, reference mesh ($R_{max} = 40$ Bohr, $N_{fe} = 16$), and atom range $Z = 1$ –92 (for Hartree–Fock, neutral closed-shell atoms only), with discretization trends consistent with the PBE panels in Figure 2. Together with the mesh construction and FeAtom comparison recorded above, these results provide additional numerical evidence for the all-electron convergence picture presented in the main paper.

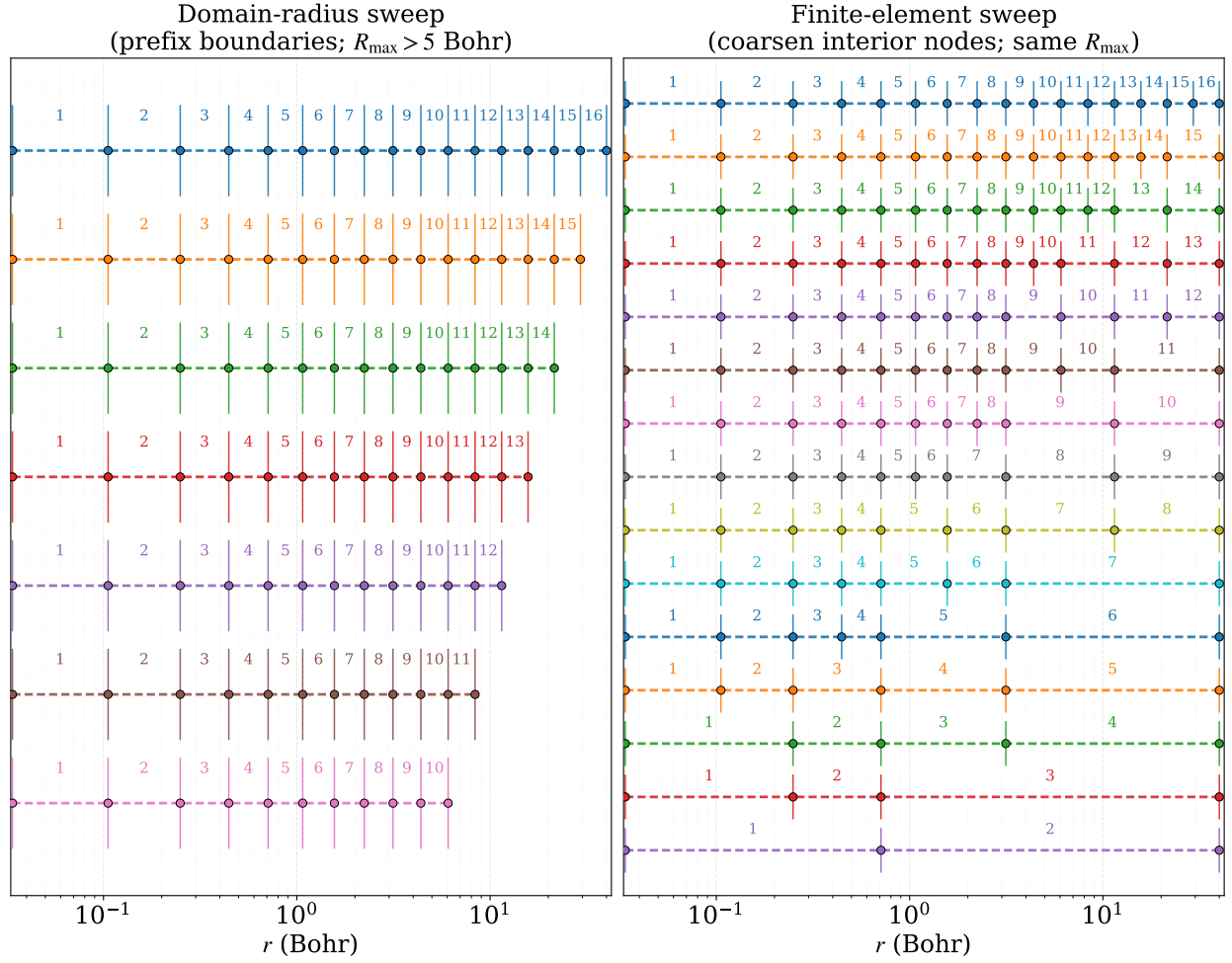


Figure 3: Nested radial mesh subsets used in all-electron convergence sweeps: each coarser level retains node locations inherited from the reference exponential mesh, so only resolution changes—not the global mesh law.

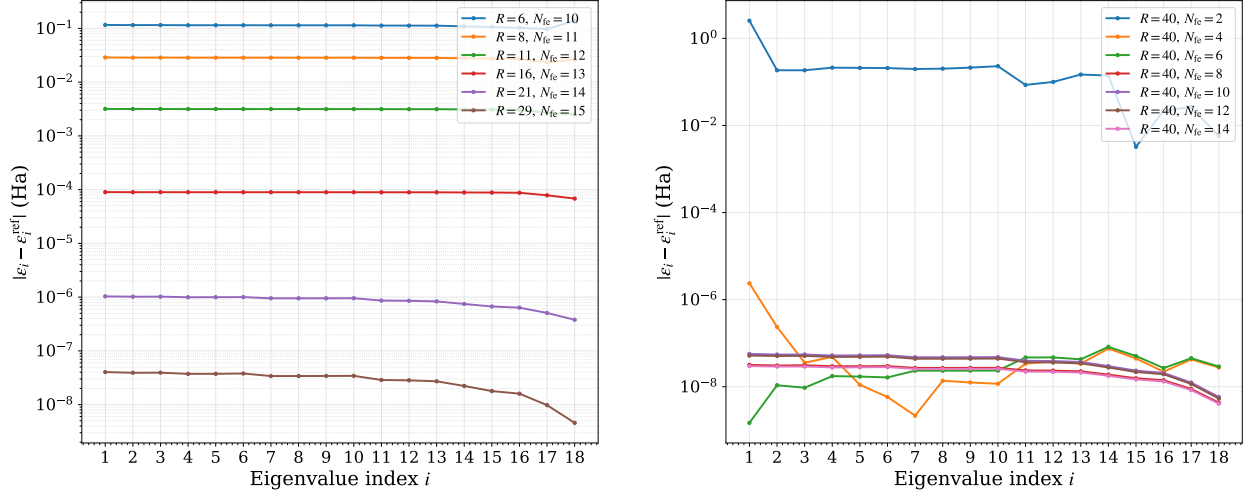


Figure 4: LDA-SVWN all-electron convergence for $Z = 92$ against FeAtom reference occupied eigenvalues at $R_{\text{max}} = 40$ Bohr and $N_{\text{fe}} = 16$: absolute eigenvalue error versus R_{max} (left) and versus N_{fe} (right).

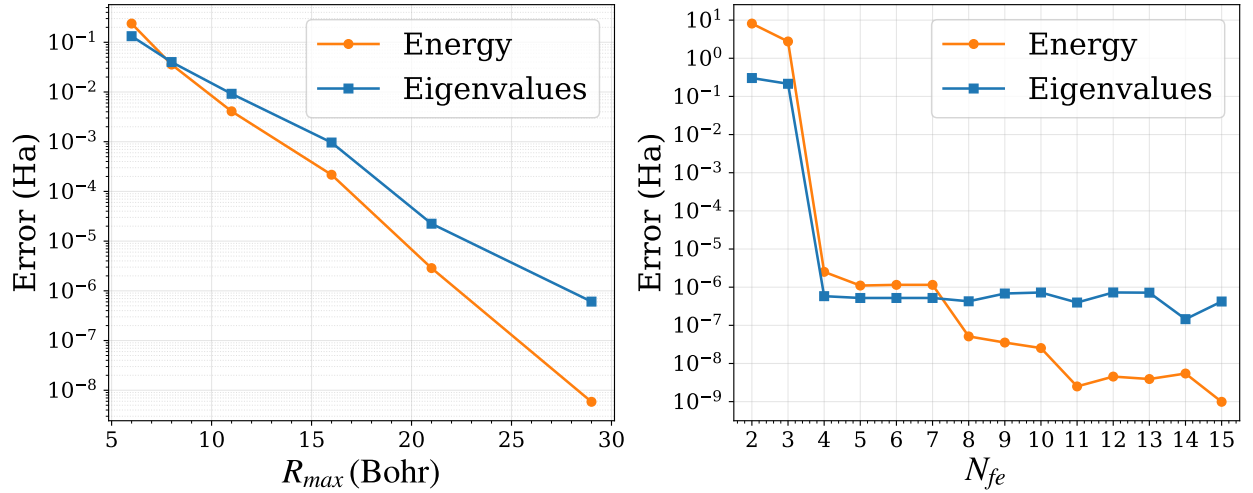


Figure 5: Same as Figure 2, for the LDA-SVWN exchange-correlation functional.

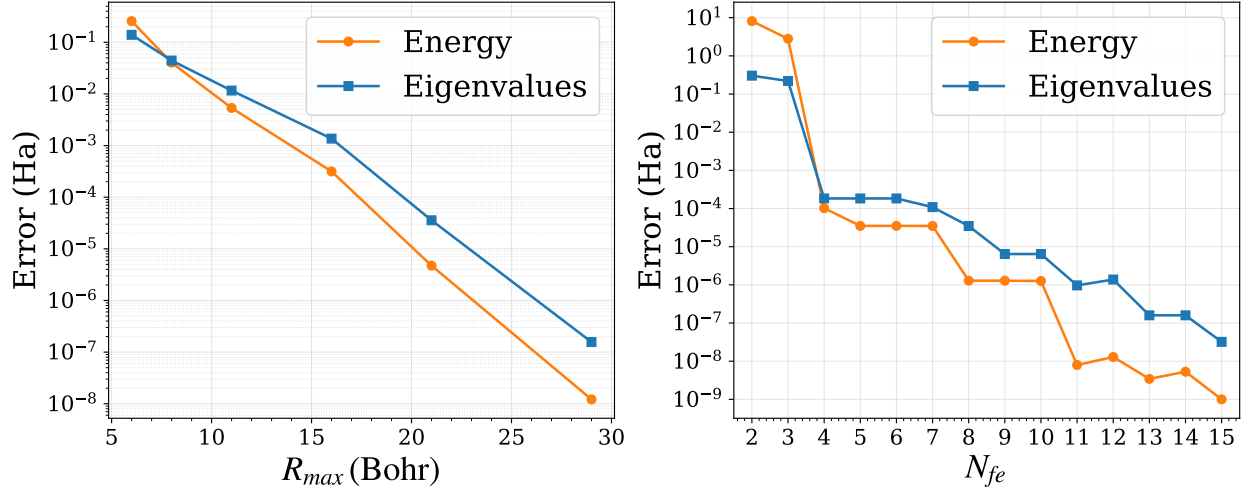


Figure 6: Same as Figure 2, for the rSCAN meta-GGA exchange-correlation functional.

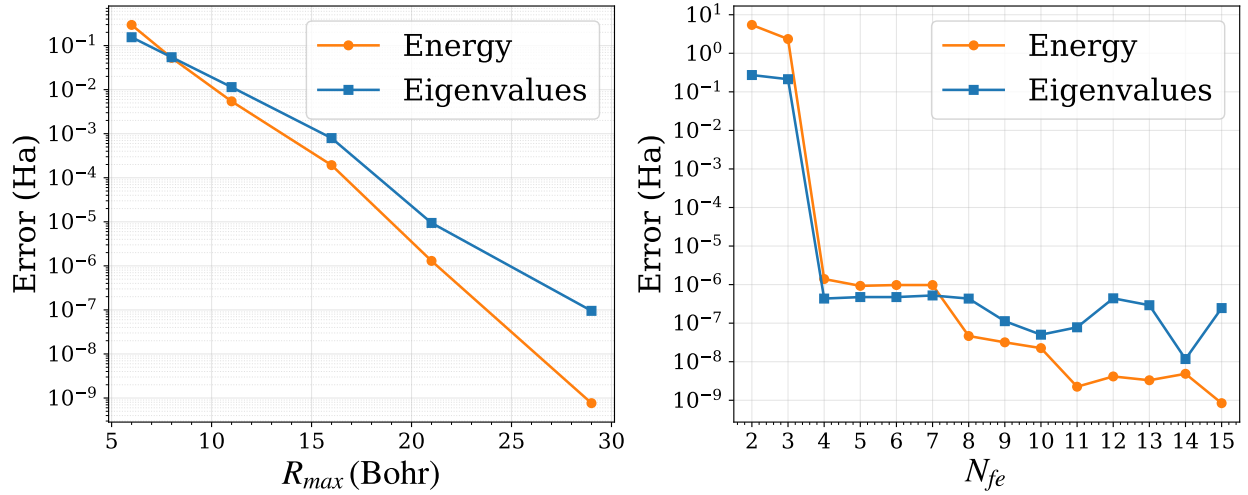


Figure 7: Same as Figure 2, for Hartree-Fock (neutral closed-shell atoms only).

3 Pseudopotential Kohn–Sham discretization convergence

We now verify the accuracy and performance of SPARC-atomSFE for pseudopotential Kohn–Sham DFT calculations. Unless otherwise specified, we use an exponential mesh with shift $c = 0$, concentration parameter $a = 20$, domain size $R_{max} = 40$ Bohr, $N_{fe} = 10$ finite elements, polynomial degree $p = 20$, and quadrature order $q = 60$. We employ the SPMS suite of ONCV pseudopotentials, which include nonlinear core corrections.

We begin by studying convergence with respect to the domain size R_{max} and the number of finite elements N_{fe} . In particular, Figure 8 shows the convergence of the total energy and occupied valence eigenvalues for the PBE functional across atoms with atomic numbers $Z = 1\text{--}57$ and $Z = 72\text{--}83$, with reference results computed using $R_{max} = 40$ Bohr and $N_{fe} = 16$. We observe that both quantities converge exponentially with increasing R_{max} , with errors below 10^{-6} Ha achieved at $R_{max} \sim 27$ Bohr. We also observe rapid convergence with increasing N_{fe} , with accuracies better than 10^{-6} Ha attained at $N_{fe} \sim 9$. These results are consistent with previous GGA results obtained using a spectral scheme based on Chebyshev polynomials. We also find similar convergence behavior for all other exchange–correlation functionals considered here, with the exception of RPA-OEP, which has not been tested in the pseudopotential context.

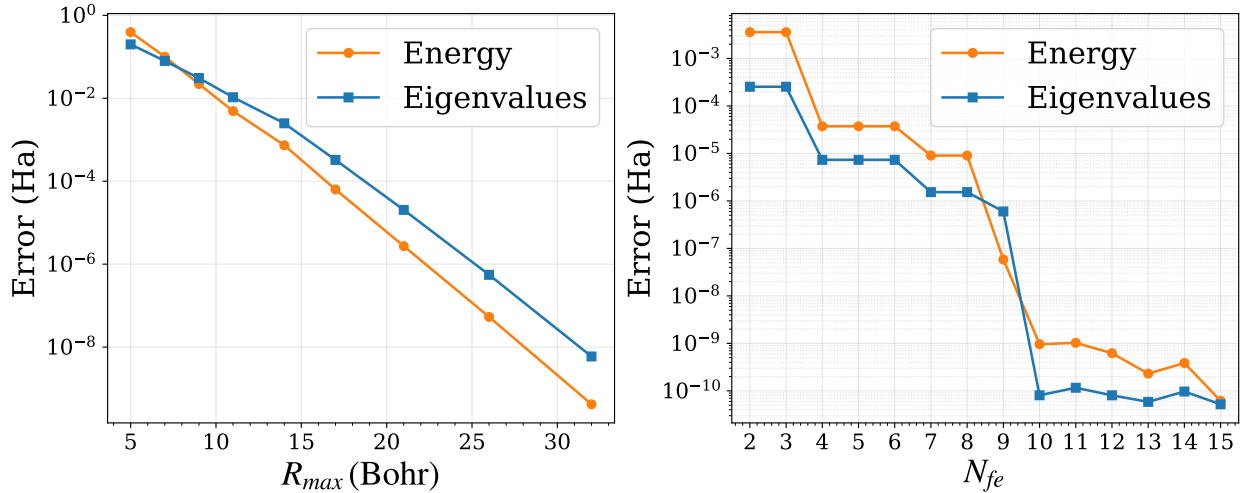


Figure 8: Convergence of the total energy and occupied valence eigenvalues with respect to domain size R_{max} (left) and number of finite elements N_{fe} (right) in pseudopotential Kohn–Sham DFT calculations, using the PBE exchange–correlation functional, for atomic numbers $Z = 1\text{--}57$ and $Z = 72\text{--}83$. The energy error is defined as the maximum absolute total-energy difference over all atoms in the set, while the eigenvalue error is defined as the maximum, over those atoms, of the per-atom mean absolute difference in the occupied valence eigenvalues. Reference values are computed using domain size $R_{max} = 40$ Bohr and $N_{fe} = 16$ finite elements.

As supplementary material to the convergence discussion presented in the main paper, the pseudopotential sweeps likewise use *nested* subsets of a common reference exponential mesh, rather than regenerating independent meshes for each choice of R_{max} or N_{fe} . This construction isolates the effect of radial resolution while preserving the underlying exponential mesh distribution across discretization levels. Figure 9 illustrates this subset construction for the pseudopotential case.

Finally, Figures 10, 11, and 12 give LDA-SVWN, rSCAN, and PBE0 counterparts to Figure 8: the same two-panel layout, error definitions, reference mesh ($R_{max} = 40$ Bohr, $N_{fe} = 16$), and

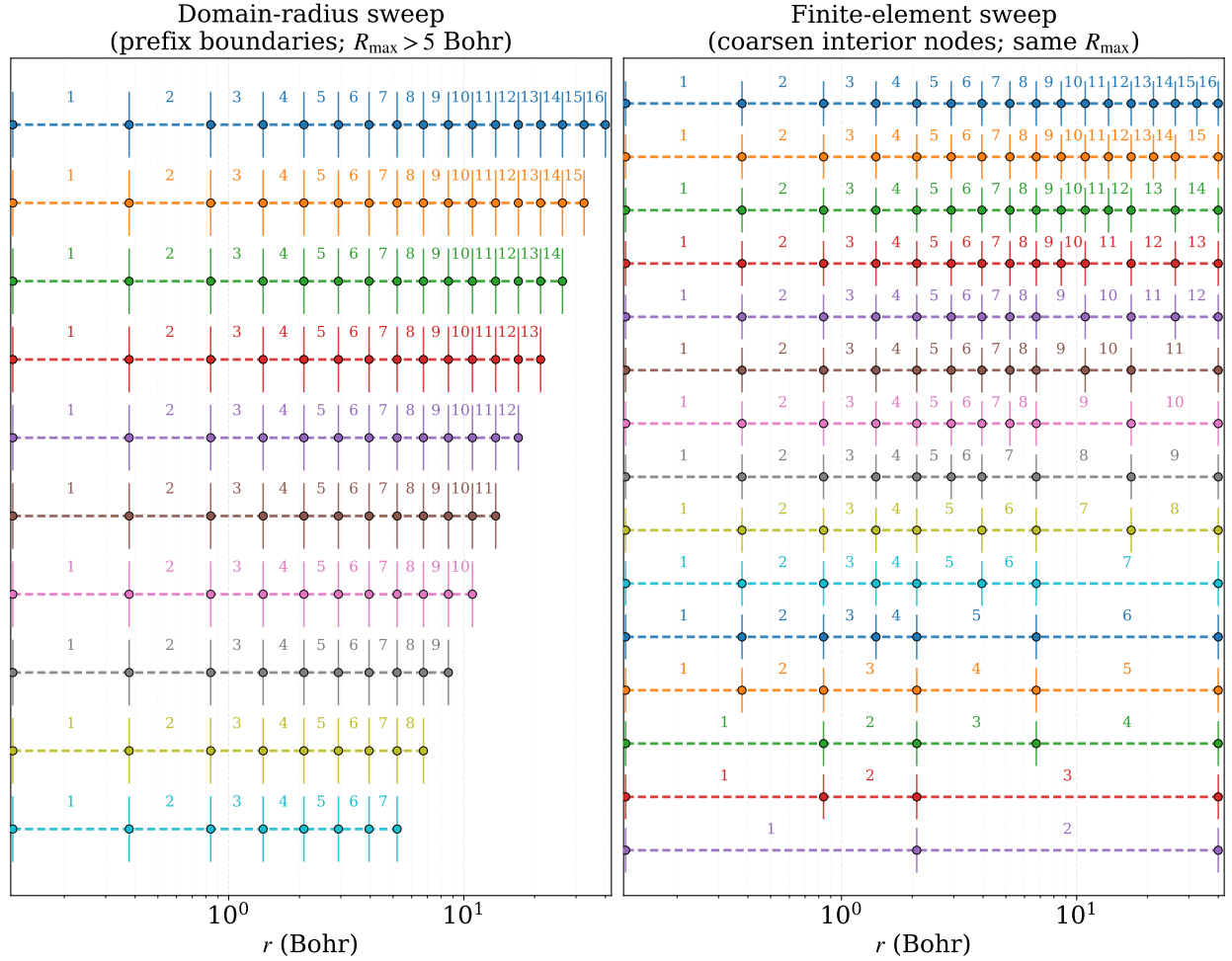


Figure 9: Nested radial mesh subsets used in pseudopotential convergence sweeps: each coarser level retains node locations inherited from the reference exponential mesh, so only resolution changes—not the global mesh law.

atom ranges $Z = 1-57$ and $Z = 72-83$, with discretization trends consistent with the PBE panels in Figure 8. Together with the mesh construction recorded above, these results provide additional numerical evidence for the pseudopotential convergence picture presented in the main paper.

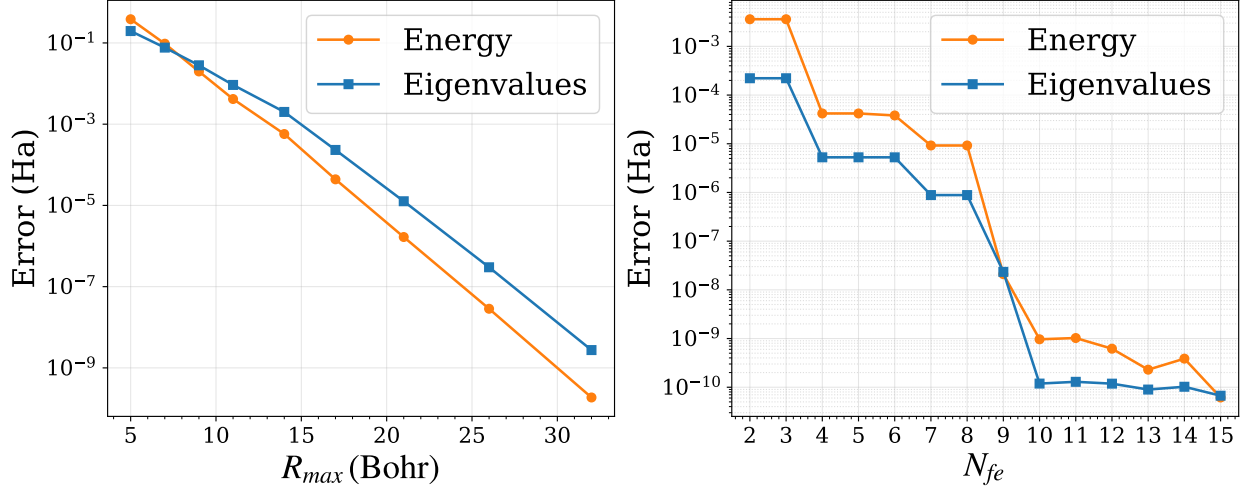


Figure 10: Same as Figure 8, for the LDA-SVWN exchange-correlation functional.

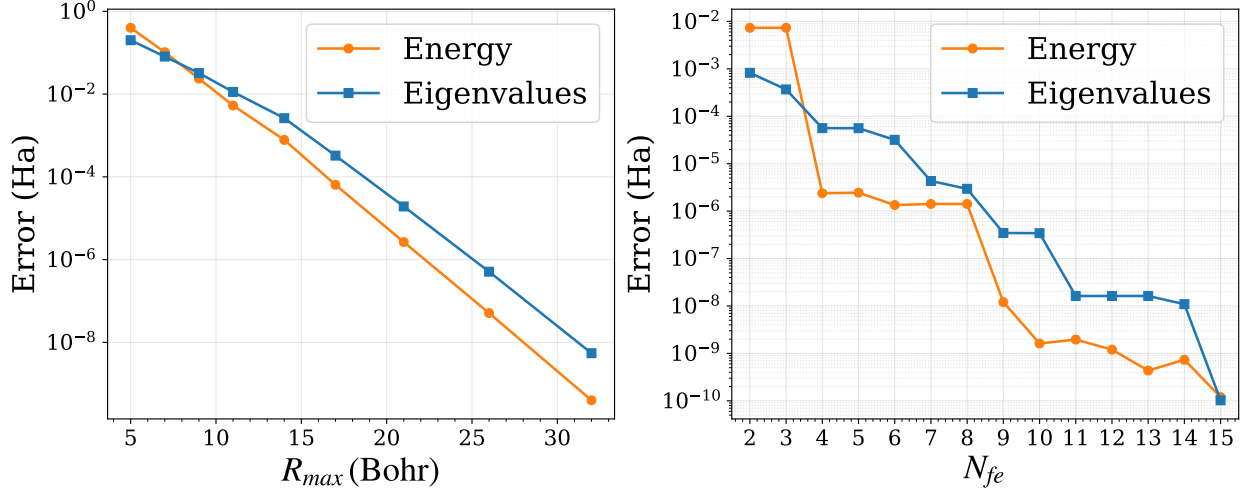


Figure 11: Same as Figure 8, for the rSCAN meta-GGA exchange-correlation functional.

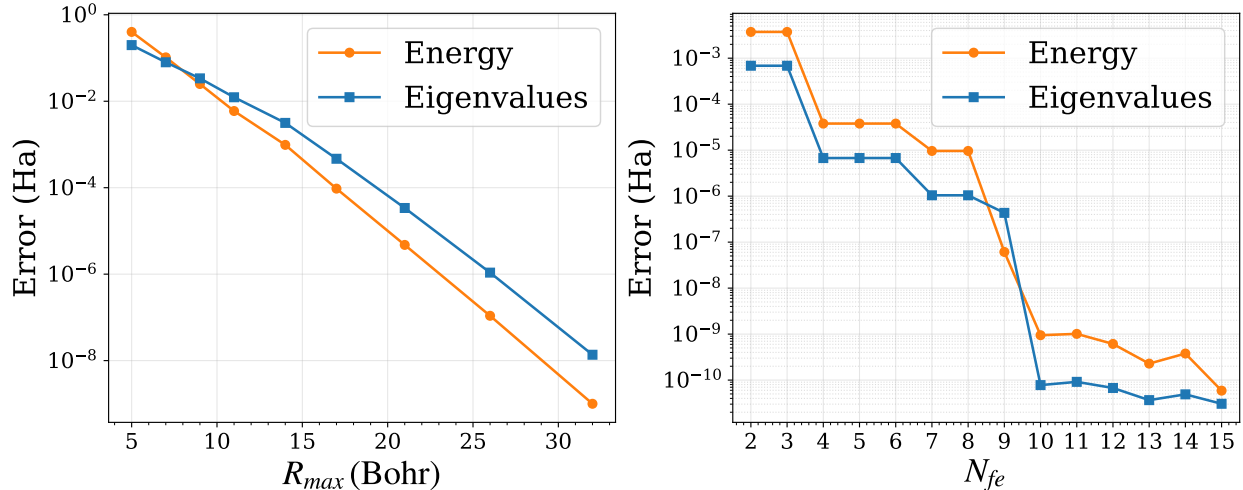


Figure 12: Same as Figure 8, for the PBE0 hybrid exchange-correlation functional.

4 All-electron Kohn–Sham accuracy

We next assess the accuracy of SPARC-atomSFE for all-electron Kohn–Sham DFT calculations through comparisons with literature codes. In particular, we consider four exchange-correlation functionals: LDA, PBE, rSCAN, and HF. For LDA, PBE, and rSCAN, results are compared for ten atoms ranging from light to heavy—H, Be, C, Ne, Na, Si, Fe, Kr, Gd, and U—against the FeAtom code for LDA and the atomPAW code for PBE and rSCAN. For HF, comparisons are carried out for closed-shell neutral atoms against Lehtola *et al.*, and for charged species—anions H^- , Li^- , F^- , Na^- , Cl^- and cations Li^+ , B^+ , Na^+ , Al^+ —against Lehtola *et al.* The distribution of the differences in the results is shown as violin plots in Figure 13. We observe that total energies and occupied eigenvalues agree to 10^{-6} Ha or better in most cases; the somewhat larger errors observed for PBE and rSCAN can likely be attributed to the accuracy of the atomPAW reference results (as we were unable to converge them further) rather than to SPARC-atomSFE itself, with rSCAN showing slightly worse agreement than PBE. Notably, for LDA, the total energy and eigenvalues of SPARC-atomSFE and FeAtom agree to 10^{-8} Ha and 10^{-9} Ha, respectively, even for the heaviest atom considered, U ($Z = 92$). These results demonstrate the accuracy of SPARC-atomSFE for atomic structure calculations based on all-electron Kohn–Sham DFT.

We also assess the accuracy of SPARC-atomSFE for all-electron RPA-OEP calculations, considering the atoms He, Be, and Ne. We employ a polynomial mesh with $s = 2$, $R_{\max} = 13$ bohr, $p_1 = 15$, $p_3 = 4$, and $q = 55$; N_{fe} , $l_{\max}$, and N_{ω} are (20, 18, 30), (30, 18, 40), and (30, 40, 60) for He, Be, and Ne, respectively, which converges energies and eigenvalues to within 5×10^{-4} Ha. The results agree with the literature to within 10^{-3} Ha, which is the accuracy typically sought/attained in such calculations.

As supplementary material to the accuracy discussion above, we tabulate the full numerical data underlying Figure 13 in Tables 2, 3, 4, 5, and 6 (LDA-SVWN, PBE, rSCAN, Hartree–Fock closed-shell, and Hartree–Fock charged species, respectively).

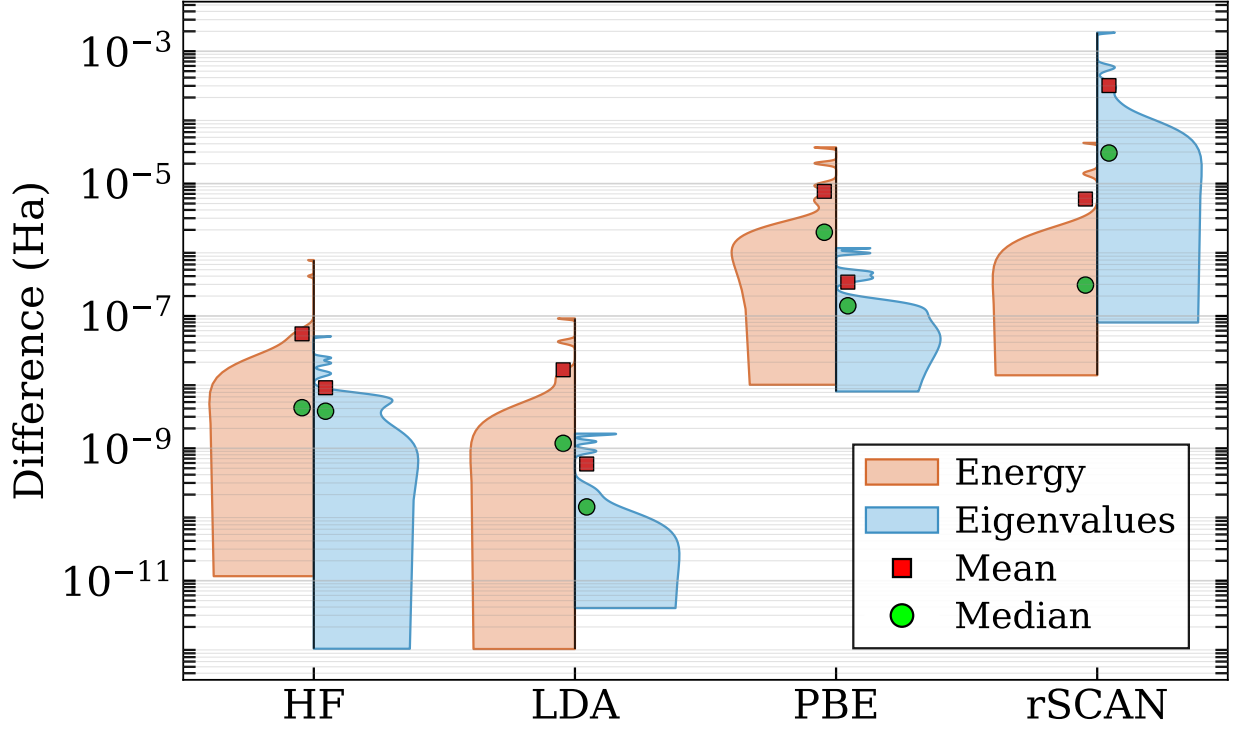


Figure 13: Comparison of SPARC-atomSFE all-electron Kohn-Sham DFT results with literature for various exchange-correlation functionals. The energy and eigenvalue errors denote the absolute differences in the corresponding quantities.

Table 2: Comparison of total energies and eigenvalues for the choice of LDA-SVWN exchange-correlation functional.

Z	$-E_{\text{tot}}$ (Ha)			Occupied-eigenvalue Difference (Ha)	
	atomSFE	FeAtom	Difference	Max $ \Delta\varepsilon $	Mean $ \Delta\varepsilon $
H 1	0.44567051824	0.44567051824	9.29×10^{-13}	3.86×10^{-12}	3.86×10^{-12}
Be 4	14.4472094741	14.4472094740	8.31×10^{-11}	3.10×10^{-11}	1.90×10^{-11}
C 6	37.4257485361	37.4257485360	1.98×10^{-10}	9.68×10^{-11}	4.20×10^{-11}
Ne 10	128.233481270	128.233481269	7.44×10^{-10}	3.64×10^{-10}	2.17×10^{-10}
Na 11	161.440060310	161.440060309	9.04×10^{-10}	6.99×10^{-11}	4.39×10^{-11}
Si 14	288.198396604	288.198396603	1.48×10^{-09}	1.37×10^{-11}	8.03×10^{-12}
Fe 26	1261.09305585	1261.09305584	5.67×10^{-09}	2.04×10^{-09}	1.26×10^{-09}
Kr 36	2750.14794043	2750.14794042	1.22×10^{-08}	1.50×10^{-09}	9.08×10^{-10}
Gd 64	10816.6538772	10816.6538772	4.09×10^{-08}	2.37×10^{-09}	1.67×10^{-09}
U 92	25658.4178889	25658.4178888	9.18×10^{-08}	2.67×10^{-09}	1.63×10^{-09}

Table 3: Comparison of total energies and eigenvalues for the choice of PBE exchange-correlation functional.

	<i>Z</i>	$-E_{\text{tot}}$ (Ha)			Occupied-eigenvalue Difference (Ha)	
		atomSFE	atomPAW	Difference	Max $ \Delta\varepsilon $	Mean $ \Delta\varepsilon $
H	1	0.458928588	0.458928597	9.13×10^{-09}	7.22×10^{-09}	7.22×10^{-09}
Be	4	14.62994793	14.62994772	2.14×10^{-07}	9.80×10^{-08}	5.27×10^{-08}
C	6	37.74820924	37.74820878	4.55×10^{-07}	4.04×10^{-08}	2.95×10^{-08}
Ne	10	128.8664291	128.8664277	1.38×10^{-06}	1.34×10^{-07}	9.12×10^{-08}
Na	11	162.1645960	162.1645945	1.55×10^{-06}	2.51×10^{-07}	1.32×10^{-07}
Si	14	289.2027550	289.2027529	2.13×10^{-06}	5.84×10^{-07}	1.53×10^{-07}
Fe	26	1263.295438	1263.295432	5.63×10^{-06}	2.21×10^{-06}	3.70×10^{-07}
Kr	36	2753.416118	2753.416109	9.33×10^{-06}	1.37×10^{-06}	4.58×10^{-07}
Gd	64	10823.19862	10823.19860	2.03×10^{-05}	8.75×10^{-06}	1.06×10^{-06}
U	92	25668.41171	25668.41167	3.53×10^{-05}	1.05×10^{-05}	8.92×10^{-07}

Table 4: Comparison of total energies and eigenvalues for the choice of rSCAN exchange-correlation functional.

	<i>Z</i>	$-E_{\text{tot}}$ (Ha)			Occupied-eigenvalue Difference (Ha)	
		atomSFE	atomPAW	Difference	Max $ \Delta\varepsilon $	Mean $ \Delta\varepsilon $
H	1	0.452860365	0.452860352	1.27×10^{-08}	7.97×10^{-08}	7.97×10^{-08}
Be	4	14.65119802	14.65119794	8.55×10^{-08}	8.66×10^{-06}	4.68×10^{-06}
C	6	37.76635704	37.76635699	5.21×10^{-08}	1.34×10^{-05}	5.75×10^{-06}
Ne	10	128.9723925	128.9723925	4.62×10^{-08}	5.24×10^{-05}	2.29×10^{-05}
Na	11	162.2862159	162.2862160	6.15×10^{-08}	6.51×10^{-05}	2.19×10^{-05}
Si	14	289.3767993	289.3767988	5.01×10^{-07}	1.22×10^{-04}	3.51×10^{-05}
Fe	26	1263.668535	1263.668536	1.04×10^{-06}	6.72×10^{-04}	1.53×10^{-04}
Kr	36	2754.194549	2754.194547	1.16×10^{-06}	1.39×10^{-03}	2.97×10^{-04}
Gd	64	10824.64115	10824.64117	1.42×10^{-05}	3.08×10^{-03}	5.69×10^{-04}
U	92	25670.94593	25670.94589	4.13×10^{-05}	1.42×10^{-02}	1.92×10^{-03}

Table 5: Hartree–Fock closed-subshell benchmark for 17 atoms: SPARC-atomSFE vs. Lehtola *et al.* Entries are total energy E_{tot} , HOMO eigenvalue $\varepsilon_{\text{HOMO}}$, and exchange energy E_x (Ha).

	<i>Z</i>	$-E_{\text{tot}}$ (Ha)			$-\varepsilon_{\text{HOMO}}$ (Ha)			$-E_x$ (Ha)		
		atomSFE	Lehtola	Difference	atomSFE	Lehtola	Difference	atomSFE	Lehtola	Difference
He	2	2.861679995600	2.861679995612	1.17×10^{-11}	0.91795556285	0.91795556286	2.57×10^{-12}	1.025768869910	1.025768869900	1.04×10^{-11}
Be	4	14.57302316828	14.57302316832	3.60×10^{-11}	0.30926955153	0.30926955157	4.15×10^{-11}	2.666913669350	2.666913669390	3.96×10^{-11}
Ne	10	128.5470981088	128.5470981094	5.79×10^{-10}	0.85040965035	0.85040965035	9.37×10^{-13}	12.10835073105	12.10835073117	1.15×10^{-10}
Mg	12	199.6146364237	199.6146364245	7.70×10^{-10}	0.25305258176	0.25305258201	2.51×10^{-10}	15.99429168655	15.99429168694	3.91×10^{-10}
Ar	18	526.8175128010	526.8175128027	1.76×10^{-9}	0.59101740571	0.59101740935	3.64×10^{-9}	30.18494199109	30.18494198731	3.78×10^{-9}
Ca	20	676.7581859224	676.7581859245	2.12×10^{-9}	0.19552969154	0.19552969247	9.30×10^{-10}	35.21120850330	35.21120850517	1.87×10^{-9}
Zn	30	1777.848116185	1777.848116191	6.10×10^{-9}	0.29250714776	0.29250714640	1.36×10^{-9}	69.64119795154	69.64119795404	2.50×10^{-9}
Kr	36	2752.054977337	2752.054977346	8.62×10^{-9}	0.52418666755	0.52418668103	1.35×10^{-8}	93.85599601220	93.85599600354	8.66×10^{-9}
Sr	38	3131.545686429	3131.545686439	9.30×10^{-9}	0.17845644076	0.17845644685	6.09×10^{-9}	101.9500832606	101.9500832705	9.84×10^{-9}
Pd	46	4937.921024056	4937.921024070	1.44×10^{-8}	0.33600200815	0.33600201367	5.52×10^{-9}	139.1429137365	139.1429137267	9.78×10^{-9}
Cd	48	5465.133142514	5465.133142530	1.57×10^{-8}	0.26485590698	0.26485591255	5.58×10^{-9}	148.9142513282	148.9142513291	9.70×10^{-10}
Xe	54	7232.138363852	7232.138363872	1.95×10^{-8}	0.45729006132	0.45729005785	3.47×10^{-9}	179.0971093189	179.0971093669	4.80×10^{-8}
Ba	56	7883.543827310	7883.543827330	2.09×10^{-8}	0.15752769303	0.15752769864	5.61×10^{-9}	189.1002498282	189.1002498415	1.33×10^{-8}
Yb	70	13391.45619308	13391.45619312	3.60×10^{-8}	0.18246254162	0.18246252223	1.94×10^{-8}	276.2138392544	276.2138392265	2.79×10^{-8}
Hg	80	18408.99149490	18408.99149494	4.50×10^{-8}	0.26104596502	0.26104601419	4.92×10^{-8}	345.3035177000	345.3035163829	1.32×10^{-6}
Rn	86	21866.77224082	21866.77224087	5.30×10^{-8}	0.42800677179	0.42800676968	2.12×10^{-9}	387.5037738100	387.5037737909	1.91×10^{-8}
Ra	88	23094.30366637	23094.30366642	5.50×10^{-8}	0.14877117348	0.14877119673	2.33×10^{-8}	401.4148473300	401.4148474301	1.00×10^{-7}

Table 6: Hartree–Fock charged benchmark: SPARC-atomSFE vs. Lehtola *et al.* Values are reported as $-E_{\text{tot}}$ (Ha). Since the reference table is reported to 9 digits after the decimal point, the absolute total-energy errors are computed from these rounded values and reported with conservative precision.

	Z	N_e	atomSFE	Lehtola	Difference
H ⁻	1	2	0.487929734	0.487929734	$< 1 \times 10^{-9}$
Li ⁺	3	2	7.236415201	7.236415201	$< 1 \times 10^{-9}$
Li ⁻	3	4	7.428231641	7.428232061	4×10^{-7}
B ⁺	5	4	24.237575184	24.237575184	$< 1 \times 10^{-9}$
F ⁻	9	10	99.459453912	99.459453913	1×10^{-9}
Na ⁺	11	10	161.676962614	161.676962614	$< 1 \times 10^{-9}$
Na ⁻	11	12	161.855125277	161.855125996	7×10^{-7}
Al ⁺	13	12	241.674670464	241.674670465	1×10^{-9}
Cl ⁻	17	18	459.576925266	459.576925268	2×10^{-9}

5 Pseudopotential Kohn–Sham accuracy

We next assess the accuracy of SPARC-atomSFE for pseudopotential Kohn–Sham DFT calculations through comparisons with the literature. In particular, we compare against the spectral scheme based SPARC-atom code, implemented as part of the M-SPARC package, while considering four exchange–correlation functionals: LDA, PBE, rSCAN, and PBE0. We compare the results for seven atoms spanning a range of chemical environments—He, N, O, Fe, Mn, Mo, and Cs. The distribution of the differences in the results is shown as violin plots in Figure 14. We observe that total energies and occupied valence eigenvalues agree to 10^{-6} Ha or better for LDA, PBE, and rSCAN. The exception is PBE0, which exhibits errors approximately an order of magnitude larger; this can be attributed to the fact that the quadrature in the spectral scheme used by SPARC-atom is limited to the degree of the polynomial used, making it difficult to converge calculations involving exact exchange, where higher-order quadrature is required. This limitation can be overcome by adopting a differential approach for the exact exchange operator, as done in the present work. These results demonstrate the accuracy of SPARC-atomSFE for atomic structure calculations based on pseudopotential Kohn–Sham DFT.

As supplementary material to the accuracy discussion above, we tabulate the full numerical data underlying Figure 14 in Tables 7, 8, 9, and 10 (LDA-SVWN, GGA-PBE, rSCAN, and PBE0, respectively).

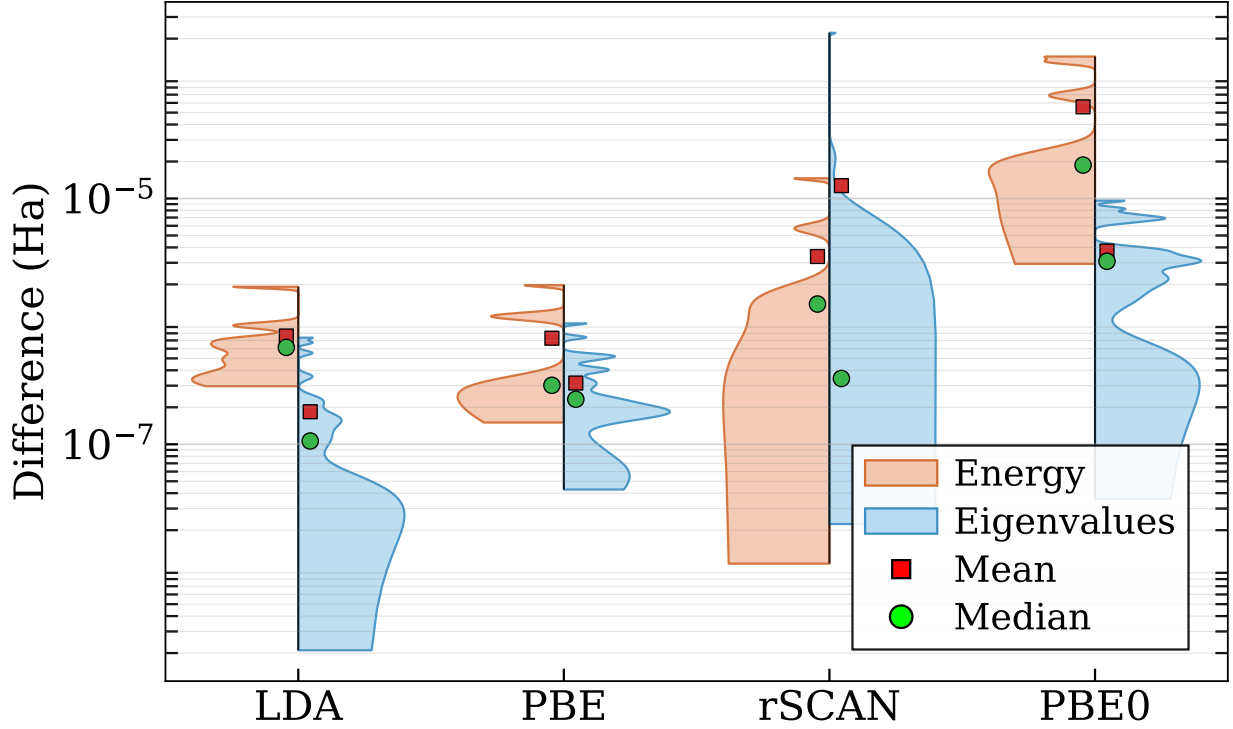


Figure 14: Comparison of SPARC-atomSFE pseudopotential Kohn-Sham DFT results with the M-SPARC SPARC-atom reference for various exchange-correlation functionals. The energy and eigenvalue errors denote the absolute differences in the corresponding quantities.

Table 7: Comparison of the total energies and eigenvalues for the LDA-SVWN exchange-correlation functional based on norm-conserving pseudopotential atomic calculations.

Z		$-E_{\text{tot}}$ (Ha)			n	ℓ	g_{nl}	$-\varepsilon_{nl}$ (Ha)		
		atomSFE	M-SPARC	Difference				atomSFE	M-SPARC	Difference
He	2	2.800084291	2.800083994	2.97×10^{-7}	1	0	2	0.57046	0.57046	1.52×10^{-7}
N	7	10.0228671	10.02286518	1.91×10^{-6}	2	0	2	0.67708	0.677081	6.72×10^{-7}
					2	1	3	0.266124	0.266124	5.53×10^{-7}
O	8	16.21748133	16.21748084	4.91×10^{-7}	2	0	2	0.873064	0.873065	7.37×10^{-7}
					2	1	4	0.338133	0.338133	3.56×10^{-7}
Mn	25	105.2337256	105.2337263	7.14×10^{-7}	3	0	2	3.138755	3.138755	4.54×10^{-8}
					3	1	6	2.002629	2.002629	3.21×10^{-9}
					3	2	5	0.257732	0.257732	2.12×10^{-7}
					4	0	2	0.194097	0.194097	1.39×10^{-8}
Fe	26	116.7411744	116.741174	3.69×10^{-7}	3	0	2	3.436694	3.436694	3.18×10^{-8}
					3	1	6	2.202574	2.202574	2.82×10^{-8}
					3	2	6	0.285108	0.285108	2.5×10^{-8}
					4	0	2	0.201428	0.201428	3.84×10^{-8}
Mo	42	69.69080029	69.69080091	6.14×10^{-7}	4	0	2	2.367637	2.367637	1.04×10^{-7}
					4	1	6	1.418245	1.418244	1.7×10^{-7}
					4	2	5	0.144562	0.144562	1.09×10^{-7}
					5	0	1	0.158758	0.158758	3.55×10^{-8}
Cs	55	23.31026826	23.31026919	9.25×10^{-7}	5	0	2	0.988689	0.988689	2.37×10^{-7}
					5	1	6	0.500169	0.500169	1.55×10^{-7}
					6	0	1	0.081715	0.081715	2.11×10^{-9}

Table 8: Comparison of the total energies and eigenvalues for the GGA-PBE exchange-correlation functional based on norm-conserving pseudopotential atomic calculations.

	Z	$-E_{\text{tot}}$ (Ha)			n	ℓ	g_{nl}	$-\varepsilon_{nl}$ (Ha)		
		atomSFE	M-SPARC	Difference				atomSFE	M-SPARC	Difference
He	2	2.853330662	2.853330926	2.64×10^{-7}	1	0	2	0.579311	0.579311	2.35×10^{-7}
N	7	10.05930315	10.05930012	3.03×10^{-6}	2	0	2	0.682914	0.682914	3.92×10^{-7}
					2	1	3	0.260551	0.260551	6.63×10^{-7}
O	8	16.27370467	16.27370633	1.66×10^{-6}	2	0	2	0.880576	0.880576	6.52×10^{-7}
					2	1	4	0.331872	0.331872	2.07×10^{-7}
Mn	25	105.3660140	105.3660158	1.83×10^{-6}	3	0	2	3.156010	3.156010	1.38×10^{-7}
					3	1	6	2.006265	2.006265	9.34×10^{-8}
					3	2	5	0.248834	0.248834	3.64×10^{-7}
Fe	26	116.7788402	116.7788403	9.29×10^{-8}	4	0	2	0.187686	0.187687	4.18×10^{-7}
					3	0	2	3.455077	3.455077	1.76×10^{-9}
					3	1	6	2.206535	2.206535	8.20×10^{-8}
					3	2	6	0.275801	0.275801	3.47×10^{-9}
Mo	42	69.67943092	69.67942912	1.79×10^{-6}	4	0	2	0.194482	0.194482	2.37×10^{-8}
					4	0	2	2.364713	2.364713	1.15×10^{-7}
					4	1	6	1.414321	1.414321	1.21×10^{-8}
					4	2	5	0.137922	0.137922	1.82×10^{-7}
Cs	55	23.24984140	23.24984231	9.15×10^{-7}	5	0	1	0.150181	0.150181	4.06×10^{-8}
					5	0	2	0.982389	0.982389	7.55×10^{-8}
					5	1	6	0.496788	0.496788	1.59×10^{-7}
					6	0	1	0.076670	0.076669	3.29×10^{-8}

Table 9: Comparison of the total energies and eigenvalues for the rSCAN exchange-correlation functional based on norm-conserving pseudopotential atomic calculations.

	Z	$-E_{\text{tot}}$ (Ha)			n	ℓ	g_{nl}	$-\varepsilon_{nl}$ (Ha)		
		atomSFE	M-SPARC	Difference				atomSFE	M-SPARC	Difference
He	2	2.803317365	2.803317354	1.07×10^{-8}	1	0	2	0.605361	0.605361	6.61×10^{-8}
N	7	9.620390181	9.620390518	3.37×10^{-7}	2	0	2	0.729363	0.729363	1.45×10^{-7}
					2	1	3	0.26564	0.265639	3.03×10^{-7}
O	8	15.70167901	15.70168048	1.47×10^{-6}	2	0	2	0.940105	0.940105	5.79×10^{-7}
					2	1	4	0.343355	0.343355	4.7×10^{-8}
Mn	25	99.17611481	99.17611343	1.38×10^{-6}	3	0	2	3.279161	3.279161	3.62×10^{-7}
					3	1	6	2.08522	2.085219	3.09×10^{-7}
					3	2	5	0.251825	0.251825	3.86×10^{-7}
Fe	26	118.0039187	118.003913	5.71×10^{-6}	4	0	2	0.195917	0.195917	2.25×10^{-8}
					3	0	2	3.585989	3.585987	2.53×10^{-6}
					3	1	6	2.293534	2.293532	2.09×10^{-6}
					3	2	6	0.281913	0.281911	1.65×10^{-6}
Mo	42	68.16700484	68.16700474	9.92×10^{-8}	4	0	2	0.203008	0.203008	6.33×10^{-7}
					4	0	2	2.45666	2.456659	3.86×10^{-7}
					4	1	6	1.469806	1.469806	3.24×10^{-7}
					4	2	5	0.13904	0.13904	2.88×10^{-7}
Cs	55	20.17964157	20.17965618	1.46×10^{-5}	5	0	1	0.154164	0.154164	7.69×10^{-8}
					5	0	2	1.023347	1.023124	2.23×10^{-4}
					5	1	6	0.51887	0.51887	1.72×10^{-7}
					6	0	1	0.076458	0.076437	2.12×10^{-5}

Table 10: Comparison of the total energies and eigenvalues for the PBE0 exchange-correlation functional based on norm-conserving pseudopotential atomic calculations.

	Z	$-E_{\text{tot}}$ (Ha)			n	ℓ	g_{nl}	$-\varepsilon_{nl}$ (Ha)		
		atomSFE	M-SPARC	Difference				atomSFE	M-SPARC	Difference
He	2	2.858444072	2.858447011	2.94×10^{-6}	1	0	2	0.668803	0.668805	1.88×10^{-6}
N	7	9.8409288	9.840936922	8.12×10^{-6}	2	0	2	0.781191	0.781194	3.11×10^{-6}
					2	1	3	0.268917	0.268919	1.45×10^{-6}
O	8	16.01035357	16.01037228	1.87×10^{-5}	2	0	2	0.999295	0.999299	3.78×10^{-6}
					2	1	4	0.36414	0.364142	2.37×10^{-6}
Mn	25	103.7537001	103.7538296	1.3×10^{-4}	3	0	2	3.363098	3.363104	6.43×10^{-6}
					3	1	6	2.162649	2.162657	8.26×10^{-6}
					3	2	5	0.258815	0.258822	6.89×10^{-6}
					4	0	2	0.214786	0.214786	4.08×10^{-7}
Fe	26	115.7186491	115.718792	1.43×10^{-4}	3	0	2	3.665564	3.665571	6.96×10^{-6}
					3	1	6	2.363997	2.364006	9.61×10^{-6}
					3	2	6	0.298827	0.298834	7.47×10^{-6}
					4	0	2	0.222093	0.222093	3.07×10^{-7}
Mo	42	68.96799747	68.96806647	6.9×10^{-5}	4	0	2	2.526861	2.526864	3.36×10^{-6}
					4	1	6	1.53111	1.531114	3.84×10^{-6}
					4	2	5	0.144336	0.144338	2.22×10^{-6}
					5	0	1	0.147808	0.147807	3.82×10^{-7}
Cs	55	22.42005527	22.42007389	1.86×10^{-5}	5	0	2	1.066609	1.066612	2.9×10^{-6}
					5	1	6	0.551291	0.551294	3.06×10^{-6}
					6	0	1	0.076059	0.076059	3.59×10^{-8}